‘REGISTRATION TO PROGRAMMING COURSE’ SYSTEM

A Project Report Submitted in Partial Fulfillment of the Requirements

AISSCE - All India Senior School Certificate Examination 2023-2024: XII-C6

In

COMPUTER SCIENCE (083)

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INTRODUCTION OF THE PROJECT

The Student Registration to a Programming Course System is a Python-based program that allows users to manage student records in a MySQL database. It offers functionalities such as registering students, editing student details, searching for students, removing students, and viewing all records. This project aims to simplify registration information management in an educational environment. the code consists of several functions, including:

- Register Course: Allows the user to register a student with their details.
- Options Menu: Presents a menu for different actions.
- Edit Record: Allows the user to edit student details.
- Search: Provides search functionality based on different criteria.
- Remove: Deletes a student record.
- display: Displays student records in a tabular format using PrettyTable.
- Start: Controls the program flow, allowing users to continue or exit.
- see all: Displays all student records.

SYSTEM REQUIREMENTS

Recommended System Requirements
Processors: Intel® Core™ i3 processor 4300M at 2.60 GHz.
Disk space: 2 to 4 GB.
Operating systems:
Windows® 10, MACOS, and UBUNTU.
Python Versions:3. X.X or Higher.

Minimum System Requirements
Processors: Intel Atom® processor or Intel® Core™ i3 processor.
Disk space: 1 GB.
Operating systems: Windows 7 or later, MACOS, and UBUNTU.
Python Versions: 2.7.X, 3.6.X

Prerequisites before installing MySQL Connector Python
You need root or administrator privileges to perform the installation process.
You must install prettytable module.
Set up a MySQL database named PROGRAMMING_COURSE and configure the connection details.

PYTHON CODING

```
import mysql.connector as my
from prettytable import PrettyTable
import os
import platform
con=my.connect(user='root',password='tiger',host='localhost',
database="PROGRAMMING_COURSE")
cursor=con.cursor()


def RegisterCourse():
    i=input("Enter Enrollment ID of student")
    n=input("Enter Name of student")
    d=input("Enter Address of student")
    p=input("Enter Phone no. of student")
    a=input("Enter Age of student")
    l=input("Enter Programming Language Selected by Student")
    f=input("Enter Fee Status of Student- Paid or Not Paid:-")
    sql="INSERT INTO STUDENT VALUES
(%s,%s,%s,%s,%s,%s,%s)"
    L=[i,n,d,p,a,l,f]
    cursor.execute(sql,L)
    con.commit()
```

```
    print("Student Id",i,"has been registered successfully")
#-----------------------------------------------------------------------------
def OptionMenu():
    print("Enter 1 :- To Register Student in Course")
    print("Enter 2 :- To Edit Student Detail ")
    print("Enter 3 :- To Search Student information ")
    print("Enter 4 :- To Remove Student ID")
    print("Enter 5 :- To See All Records")
    e = int(input("Please Select An Option :- "))
    if(e == 1):
        RegisterCourse()
    elif (e==2):
        EditRecord()
    elif (e==3):
        Search()
    elif (e==4):
        Remove()
    elif (e==5):
        seeall()
    else:
        if(platform.system() == "Windows"):
            print(os.system('cls'))
        else:
            print(os.system('clear'))
```

```
        print("Invalid choice....Enter correct choice!!! ")
        OptionMenu()
#-----------------------------------------------------------------------
def EditRecord():
    q=input("Enrollment ID of Student whose detail you want to
Update/Edit")
    print("Enter 1 :- To Update/Correct Name of Student")
    print("Enter 2 :- To Update/Edit Address")
    print("Enter 3 :- To Update/Edit Phone Number")
    print("Enter 4 :- To Update/Edit Age")
    print("Enter 5 :- To Update/Edit Language Selected")
    print("Enter 6 :- To Update/Edit Fee Status")
    y = int(input("Please Select An Option :- "))

    if y==1:
        a=input("Enter Updated/Corrected Name")
        sql="UPDATE STUDENT SET SNAME='"+a+ "'WHERE ID='"
+q+ "'"
        cursor.execute(sql)
        con.commit()
        print("Name of Student has been updated successfully")
    elif y==2:
        a=input("Enter New Address")
```

```
        sql="UPDATE STUDENT SET SADDRESS='"+a+ "'WHERE
ID='" +q+ "'"
        cursor.execute(sql)
        con.commit()
        print("Address has been updated successfully")
    elif y==3:
        a=input("Enter New Phone Number")
        sql="UPDATE STUDENT SET SPHONE='"+a+ "'WHERE ID='"
+q+ "'"
        cursor.execute(sql)
        con.commit()
        print("Phone number has been updated successfully")
    elif y==4:
        a=input("Enter Updated Age")
        sql="UPDATE STUDENT SET SAGE='"+a+ "'WHERE ID='" +q+
"'"
        cursor.execute(sql)
        con.commit()
        print("Age of student id",q,"has been updated successfully")
    elif y==5:
        a=input("Enter Programming Language Selected By Student")
        sql="UPDATE STUDENT SET PLANG='"+a+ "'WHERE ID='"
+q+ "'"
        cursor.execute(sql)
```

```
        con.commit()
        print("Programming Language of student id",q,"has been
updated successfully")
    elif y==6:
        a=input("Enter Updated Fee Status")
        sql="UPDATE STUDENT SET FEE_STATUS='"+a+ "'WHERE
ID='" +q+ "'"
        cursor.execute(sql)
        con.commit()
        print("Fee Status of student id",q,"has been updated
successfully")
    else:
        print("Invalid choice....Enter correct choice!!! ")
        EditRecord()
#-----------------------------------------------------------------------
def Search():
    print("Select the search criteria:- ")
    print("1. Enrollment Id")
    print("2. Name")
    print("3. Address")
    print("4. Phone No.")
    print("5. Age")
    print("6. Programming Language")
    print("7. Fee Status")
```

```
h=int(input("Enter your choice:- "))
if h==1:
    m=input("Enter Enrollment Id:- ")
    l=(m,)
    sql="SELECT * FROM STUDENT WHERE ID=%s"
    cursor.execute(sql,l)
elif h==2:
    m=input("Enter Student Name:- ")
    l=(m,)
    sql="SELECT * FROM STUDENT WHERE SNAME=%s"
    cursor.execute(sql,l)
elif h==3:
    m=input("Enter Address:- ")
    l=(m,)
    sql="SELECT * FROM STUDENT WHERE SADDRESS=%s"
    cursor.execute(sql,l)
elif h==4:
    m=input("Enter phone:- ")
    l=(m,)
    sql="SELECT * FROM STUDENT WHERE SPHONE=%s"
    cursor.execute(sql,l)
elif h==5:
    m=input("Enter Age:- ")
    l=(m,)
```

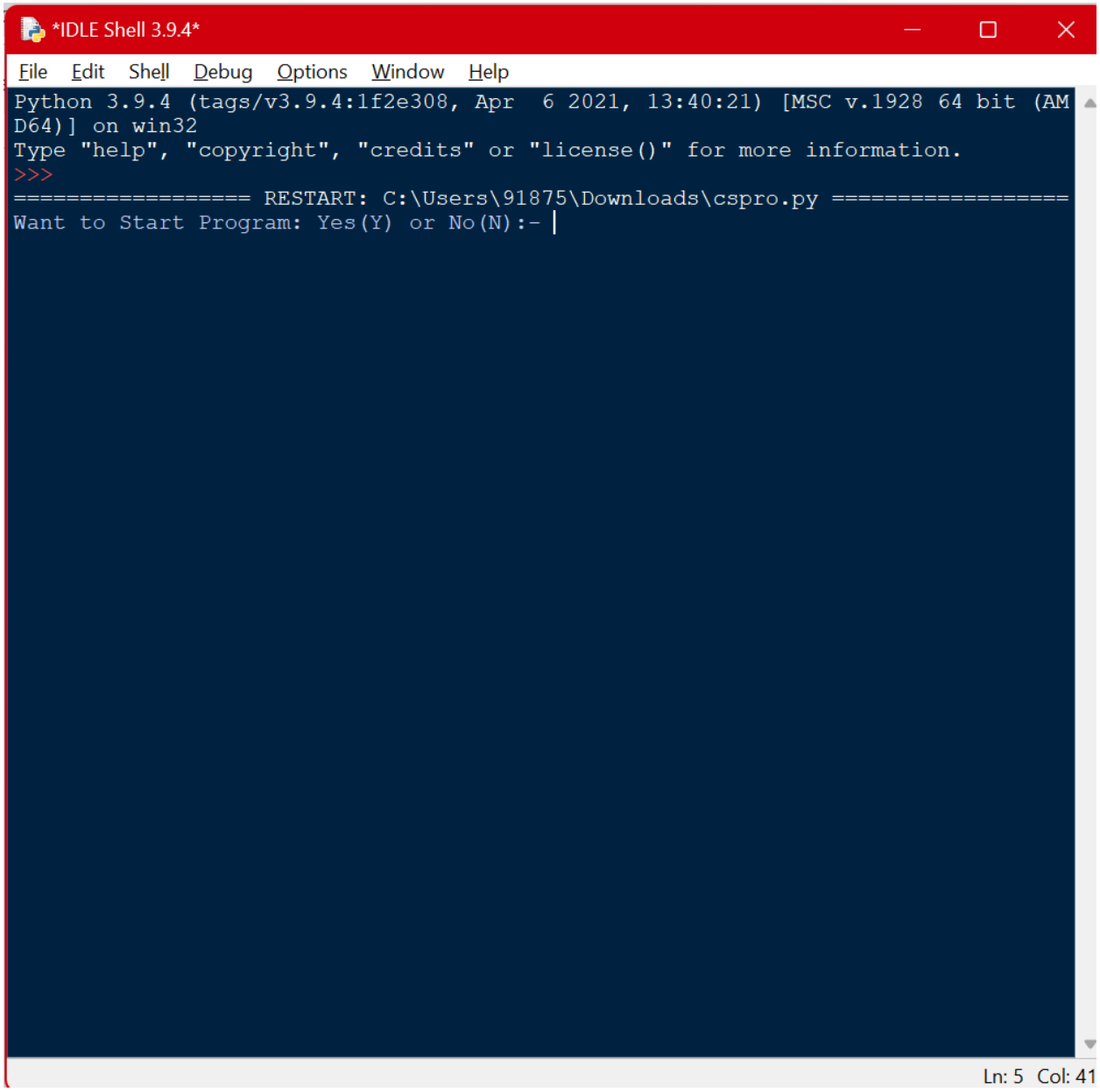
```
        sql="SELECT * FROM STUDENT WHERE SAGE=%s"
        cursor.execute(sql,l)
    elif h==6:
        m=input("Enter Programming Language:- ")
        l=(m,)
        sql="SELECT * FROM STUDENT WHERE PLANG=%s"
        cursor.execute(sql,l)
    elif h==7:
        m=input("Enter Fee Status:- ")
        l=(m,)
        sql="SELECT * FROM STUDENT WHERE FEE_STATUS=%s"
        cursor.execute(sql,l)
    else:
        print("Invalid choice....Enter correct choice!!! ")
        Search()
    result= cursor.fetchall()
    display(result)
#-----------------------------------------------------------------------
def Remove():
    r=input("Enter Enrollment Id of the Student to be deleted:- ")
    l=(r,)
    sql="DELETE FROM STUDENT WHERE ID=%s"
    cursor.execute(sql,l)
    con.commit()
```

```
    print("Record of Id=",r,"has been deleted successfully")
#-----------------------------------------------------------------------
def display(result):
    x = PrettyTable()
    x.field_names = ["Enrollment ID", "Name", "Address", "Phone
No.","Age","Programming Language","Fee Status"]
    for i in result:
        x.add_row(i)
    print(x)
#-----------------------------------------------------------------------
def Start():
    restart = input("Want to Start Program: Yes(Y) or No(N):- ")
    while(restart.lower() == 'y'):
        OptionMenu()
        print("Enter 'C' to Continue")
        print("Enter 'E' to Exit")
        j=input("Your Choice:-")
        if j.lower()=='e':
            break
        elif j.lower()=='c':
            if(platform.system() == "Windows"):
                print(os.system('cls'))
            else:
                print(os.system('clear'))
```

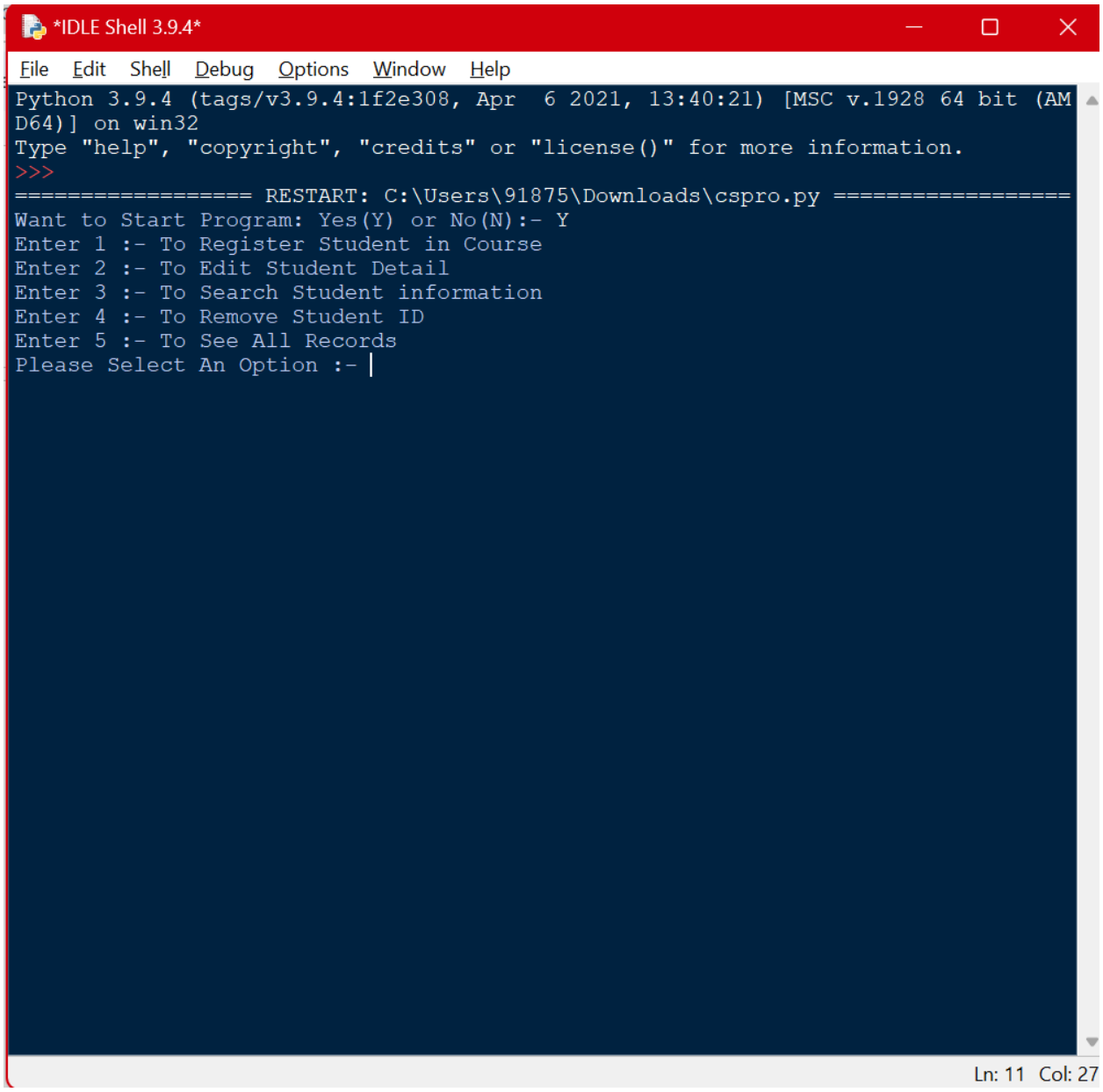
```
#-----------------------------------------------------------------------
def seeall():
    sql="SELECT * FROM STUDENT "
    cursor.execute(sql)
    r = cursor.fetchall()
    display(r)

Start()
```

OUTPUT

OPENING SCREEN



MAIN MENU


IDLE Shell 3.9.4
File Edit Shell Debug Options Window Help
Python 3.9.4 (tags/v3.9.4:1f2e308, Apr 6 2021, 13:40:21) [MSC v.1928 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:\Users\91875\Downloads\cspro.py ==========
Want to Start Program: Yes(Y) or No(N):- Y
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :-
Ln: 11 Col: 27

REGISTRATION OF STUDENT

IDLE Shell 3.9.4

File Edit Shell Debug Options Window Help

```
Python 3.9.4 (tags/v3.9.4:1f2e308, Apr  6 2021, 13:40:21) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================== RESTART: C:\Users\91875\Downloads\cspro.py ==================
Want to Start Program: Yes(Y) or No(N):- Y
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :- 1
Enter Enrollment ID of student03
Enter Name of studentSOH
Enter Address of studentTN
Enter Phone no. of student044345
Enter Age of student17
Enter Programming Language Selected by StudentPYTHON
Enter Fee Status of Student- Paid or Not Paid:-Paid
```

Ln: 13 Col: 0

EDITING DETAILS OF STUDENT

```
Enter 'E' to Exit
Your Choice:-C
0
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :- 2
Enrollment ID of Student whose detail you want to Update/Edit1
Enter 1 :- To Update/Correct Name of Student
Enter 2 :- To Update/Edit Address
Enter 3 :- To Update/Edit Phone Number
Enter 4 :- To Update/Edit Age
Enter 5 :- To Update/Edit Language Selected
Enter 6 :- To Update/Edit Fee Status
Please Select An Option :- 1
Enter Updated/Corrected Namevish
Name of Student has been updated successfully
Enter 'C' to Continue
Enter 'E' to Exit
Your Choice:-
```

SEARCHING STUDENT'S INFORMATION

```
*IDLE Shell 3.9.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.4 (tags/v3.9.4:1f2e308, Apr  6 2021, 13:40:21) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================== RESTART: C:\Users\91875\Downloads\cspro.py ==================
Want to Start Program: Yes(Y) or No(N):- Y
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :- 3
Select the search criteria:-
1. Enrollment Id
2. Name
3. Address
4. Phone No.
5. Age
6. Programming Language
7. Fee Status
Enter your choice:- 2
Enter Student Name:- MARK ANTONY
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
| Enrollment ID |     Name    | Address | Phone No. | Age | Programming Language
 | Fee Status |
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
|       2       | mark antony |  bombay |   112567  |  18 |          C
 |  Not Paid  |
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
Enter 'C' to Continue
Enter 'E' to Exit
Your Choice:-
```

DELETING STUDENT ID

```
*IDLE Shell 3.9.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.4 (tags/v3.9.4:1f2e308, Apr  6 2021, 13:40:21) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================== RESTART: C:\Users\91875\Downloads\cspro.py ==================
Want to Start Program: Yes(Y) or No(N):- Y
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :- 4
Enter Enrollment Id of the Student to be deleted:- 3
Record of Id= 3 has been deleted successfully
Enter 'C' to Continue
Enter 'E' to Exit
Your Choice:-
```

SEEING ALL RECORDS

```
*IDLE Shell 3.9.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.4 (tags/v3.9.4:1f2e308, Apr  6 2021, 13:40:21) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================== RESTART: C:\Users\91875\Downloads\cspro.py ==================
Want to Start Program: Yes(Y) or No(N):- Y
Enter 1 :- To Register Student in Course
Enter 2 :- To Edit Student Detail
Enter 3 :- To Search Student information
Enter 4 :- To Remove Student ID
Enter 5 :- To See All Records
Please Select An Option :- 5
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
| Enrollment ID |     Name    | Address | Phone No. | Age | Programming Language
 | Fee Status |
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
|       1       |     vish    |  dubai  |   44567   |  17 |        python
 |    Paid    |
|       2       | mark antony |  bombay |   112567  |  18 |          C
 |  Not Paid  |
+---------------+-------------+---------+-----------+-----+--------------------
-+------------+
Enter 'C' to Continue
Enter 'E' to Exit
Your Choice:-
```

MYSQL DATABASE AND TABLES USED IN THIS PROJECT

DATABASE&TABLE

```
MySQL 8.1 Command Line Cli

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| programming_course |
| sys                |
+--------------------+
5 rows in set (0.08 sec)

mysql> use programming_course;
Database changed
mysql> show tables;
+------------------------------+
| Tables_in_programming_course |
+------------------------------+
| student                      |
+------------------------------+
1 row in set (0.14 sec)
```

STUDENT TABLE

MySQL 8.1 Command Line Cli

```
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| programming_course |
| sys                |
+--------------------+
5 rows in set (0.08 sec)

mysql> use programming_course;
Database changed
mysql> show tables;
+------------------------------+
| Tables_in_programming_course |
+------------------------------+
| student                      |
+------------------------------+
1 row in set (0.14 sec)

mysql> select * from student;
+----+-------------+----------+--------+------+--------+------------+
| ID | SNAME       | SADDRESS | SPHONE | SAGE | PLANG  | FEE_STATUS |
+----+-------------+----------+--------+------+--------+------------+
|  1 | vish        | dubai    |  44567 |   17 | python | Paid       |
|  2 | mark antony | bombay   | 112567 |   18 | C      | Not Paid   |
|  3 | soh         | TN       |  44345 |   17 | PYTHON | Paid       |
+----+-------------+----------+--------+------+--------+------------+
3 rows in set (0.00 sec)

mysql>
```

REFERENCES

www.python.org
www.programiz.com
www.mysql.com

Thank You!